fuchs2026

Catherine Maximous

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Revised Proposal

My initial proposal was to replicate Bosker et al. (2014)'s paper on native and non-native disfluencies. However, after receiving valuable feedback and talking to Haoyu about the feasibility of the project, I decided to pursue another replication project that can fit better within the scope of the course. Due to the time constraints of having to replicate, collect data, and analyze the project within one quarter, I may not be able to create all the materials needed on time. For example, the study uses Dutch audio recordings and words while I would have

to use English to be able to recruit participants. I would also need to match the frequency, target onset times, and reaction times for each English target. Additionally, the experiment's main findings were from eye-tracking rather than the reaction time. Below is my new proposal with a paper that does a behvaioral study and has open source materials to ensure better replication.

Introduction

Recent research on language variability has demonstrated how different language environments shape language and cognitive processes. Specifically, Heritage bilinguals offer a powerful lens for understanding how becoming a dominant speaker in the second acquired language despite not having the first years of development in the language can provide insights into the plasticity of the language system. Therefore, understanding how language processing is shaped by the heritage bilingual experience and how that may differ from monolingual or second language learner is an important gap to address in the bilingualism literature.

I am proposing a new project proposal, where I would like to further explore heritage speakers' abilities to comprehend relative clauses across varying syntactic complexity. Therefore, in order to test this, I will be replicating Fuchs and colleagues (2026)'s paper where they compared heritage speakers to English-raised bilinguals who are late bilinguals and learned Spanish only in the classroom setting and Spanish-raised bilinguals who were raised in Spanish-speaking dominant environment. They tested participants' constraints when cognitive loading increased alongside syntactic complexity but also tested cognitive effort and syntactic complexity separately. For the purpose of this class, I will be replicating the experiment with heritage speakers in the picture-matching experiment which tested syntactic competence. The paper looked at three different types of sentences: Reduced Relative (RR), Subject Relative (SR), Object Relative (OR) in Spanish. RR tends to be the easiest condition as it follows a direct order of events for example "the bear painting the tiger." SR on the other hand is when the subject becomes the actor so instead it is "the bear that is painting the tiger." Finally, OR which is the most difficult syntactically as it starts with the noun where the sentence instead states "the bear that the tiger paints." In this sentence, it is typically harder for people as they often initially assume the bear is the one enacting the painting and not the tiger but then have to revise their interpretation once they hear the full sentence. The key findings of this experiment is that heritage speakers showed higher accuracy for Reduced Relatives than Subject Relatives ($= 0.87$, SE $= 0.22$, z $= 3.86$, p $< .001$) and significantly higher accuracy for Subject Relatives than Object Relatives ($= 5.07$, SE $= 0.17$, z $= 29.52$, p $< .001$), This means that heritage speakers tend to have difficulties in interpreting object relative clauses and show a relative-clause complexity effect.

Methods

To conduct this experiment, I will be implementing the experimental design from Fuchs and colleagues' (2026) design. Participants will hear a short Spanish phrase containing a relative clause and are instructed to select the picture that correctly matches the described event. Each trial will display two images depicting reversed agent-patient roles (e.g., a bear painting a tiger vs. a tiger painting a bear). The auditory stimuli will follow the structure of Article + Nouns + Relative Clause. As the task progresses, syntactic complexity will increase by using three types of relative clauses: Reduced Relatives, Subject Relative, and Object Relatives. All materials for this experiment has been provided through the OSF repository: https://osf.io/et-prk/files/osfstorage?view_only=19f67921b4eb4ab58299685068ac0a16 along with analysis materials ensuring better replication of the experiment.

The paper recruited 31 heritage Spanish speakers, who were born and raised in the U.S. or immigrated by age 6 and reported speaking Spanish with one or both parents. They also report English to be their most dominant language but having acquired Spanish since birth. Ensuring that I am recruiting the correct population and including the same criteria as the researchers will be the most important and challenging part of the project. This is because it will restrict the availability of people I can recruit from SONA but will be the most fitting to what the study tested. To do so, I will include similar language history questionnaires as the paper, the Bilingual Language Profile (BLP) and additional questions from the Language Experience and Proficiency Questionnaire (LEAP-Q). Additionally, the organization and coding of the experiment on JavaScript will also be another learning challenge since participants completed 120 picture-matching trials so it is a substantive amount of images and audios to be coded. I would also like to extend this study by including variations of heritage speakers. The study categorized heritage speakers as those who report speaking Spanish with one or more parent and were born and raised in the U.S. or immigrated at the age of 6 or younger. All of the heritage speakers' report English dominance as they experience reduced Spanish input in an English dominant environment (U.S.). However, the study does not reveal variations within the heritage speakers' themselves and how those who more production experience may differ than those with limited production abilities but tends to be better at comprehension. This way I will be testing out to what extent does proficiency and immersion in the heritage language moderate the relative-clause complexity effect that the authors report for the heritage speakers.

Learning Goals

I am hoping through this project, to learn how to build a task through jsPsych that is well structured. I have not had much experience with building new tasks especially in jsPsych however as I progress in building my second year project for the cognitive science program, I will need to design and build new tasks to fit my project aims. I also want to learn implementing similar analysis and methods in a published paper in order to implement some type

of replication within a new project. This will help ensure that I am following up and building on these results and testing whether these effects exist before beginning a whole new project. I believe this project will be feasible as all the materials have been provided but I will need to ensure that I am organizing the task correctly and starting participant recruitment early to ensure I have enough participants for analysis. Additionally, this will be my first study looking at comprehension within bilinguals rather than production which will help inform my future projects towards assessing comprehension abilities and individual differences within heritage bilinguals.

Power Analysis

####Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

The key test for this project is a group main effect (HS vs. ERB) on single-task picture-matching accuracy, from a logistic mixed-effects.

To detect the effect of subject relatives being more accurate than object relatives, the following sample size is needed for at 80%, 90%, and 95% power. The paper reported that heritage speakers (HS) were being more accurate on Reduced Relatives than on Subject Relatives. They reported a z-value of 3.86 which is a strong evidence for the effect. The original study included 31 heritage speakers. I have taken the following z-value and divided it by the square root of the number of people tested in the paper to see what is the effect size for this within-person comparison. This shows me the effect size of the original paper which is 0.69, a medium effect. To detect this effect again assuming that the effect is only half as big as the paper's then my sample size needs to be bigger than the paper. At 80%, I would need 68 people, 90 at 90% power, and 110 at 95% power. Due to limitations in time and amount of sign ups that I may receive through SONA, 68 people is the the most feasible sample size.

```
library(pwr)
```

```
Warning: package 'pwr' was built under R version 4.4.3
```

```
dz <- 3.86 / sqrt(31)
dz
```

```
[1] 0.6932765
```

```
sapply(c(.8, .9, .95), \(p) pwr.t.test(d = dz / 2, power = p, type = "paired")$n)
```

```
[1]  67.26714  89.38952 110.08759
```

```r
library(pwr)

dz_ci <- function(dz, n, level = .95) {
  se <- sqrt(1 / n + dz^2 / (2 * n))
  z <- qnorm(1 - (1 - level) / 2)
  c(lower = dz - z * se, upper = dz + z * se)
}

# ================ HS, N = 31 ================

# ---- HS: RR vs. SR, z = 3.86 ----
dz_hs_rr_sr <- 3.86 / sqrt(31)
dz_hs_rr_sr
```

```
[1] 0.6932765
```

```r
# Half the original effect
pwr.t.test(d = dz_hs_rr_sr / 2, power = .8, type = "paired")$n
```

```
[1] 67.26714
```

```r
# 2.5x the original N (people)
2.5 * 31
```

```
[1] 77.5
```

```r
# Safeguard power: lower end of a 60% CI on the original effect
dz_hs_rr_sr_safe <- dz_ci(dz_hs_rr_sr, n = 31, level = .60)["lower"]
dz_hs_rr_sr_safe
```

```
    lower
0.5249308
```

```r
pwr.t.test(d = dz_hs_rr_sr_safe, power = .8, type = "paired")$n
```

```
[1] 30.46064
```

```r
# ---- HS: SR vs. OR, z = 29.52 ----
dz_hs_sr_or <- 29.52 / sqrt(31)
dz_hs_sr_or
```

```
[1] 5.301949
```

```r
# Half the original effect
pwr.t.test(d = dz_hs_sr_or / 2, power = .8, type = "paired")$n
```

```
[1] 3.394686
```

```r
# 2.5x the original N (people)
2.5 * 31
```

```
[1] 77.5
```

```r
# Safeguard power: lower end of a 60% CI on the original effect
dz_hs_sr_or_safe <- dz_ci(dz_hs_sr_or, n = 31, level = .60)["lower"]
dz_hs_sr_or_safe
```

```
   lower
4.715431
```

```r
pwr.t.test(d = dz_hs_sr_or_safe, power = .8, type = "paired")$n
```

```
[1] 2.545201
```

```r
# ================= ERB, N = 21 =================

# ---- ERB: RR vs. SR, z = 2.14 ----
dz_erb_rr_sr <- 2.14 / sqrt(21)
dz_erb_rr_sr
```

```
[1] 0.4669863
```

```r
# Half the original effect
pwr.t.test(d = dz_erb_rr_sr / 2, power = .8, type = "paired")$n
```

```
[1] 145.8979
```

```r
# 2.5x the original N (people)
2.5 * 21
```

```
[1] 52.5
```

```r
# Safeguard power: lower end of a 60% CI on the original effect
dz_erb_rr_sr_safe <- dz_ci(dz_erb_rr_sr, n = 21, level = .60)["lower"]
dz_erb_rr_sr_safe
```

```
    lower
0.2735757
```

```r
pwr.t.test(d = dz_erb_rr_sr_safe, power = .8, type = "paired")$n
```

```
[1] 106.8067
```

```r
# ---- ERB: SR vs. OR, z = 24.12 ----
dz_erb_sr_or <- 24.12 / sqrt(21)
dz_erb_sr_or
```

```
[1] 5.263416
```

```r
# Half the original effect
pwr.t.test(d = dz_erb_sr_or / 2, power = .8, type = "paired")$n
```

```
[1] 3.411188
```

```r
# 2.5x the original N (people)
2.5 * 21
```

```
[1] 52.5
```

```r
# Safeguard power: lower end of a 60% CI on the original effect
dz_erb_sr_or_safe <- dz_ci(dz_erb_sr_or, n = 21, level = .60)["lower"]
dz_erb_sr_or_safe
```

```
   lower
4.555639
```

```r
pwr.t.test(d = dz_erb_sr_or_safe, power = .8, type = "paired")$n
```

```
[1] 2.578788
```

For 80% power, to find the SR > OR effect which is the main effect, I need at least 78 heritage speakers and if I am able to recruit ERBs which are L2 speakers, I would need a sample size of 53. Since I am not sure which population I may collect more, I am going to aim for collecting both and seeing if I can get a big enough sample size to see a true effect.

4. Make a power curve

```r
library(pwr)
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.4.3

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'tibble' was built under R version 4.4.3

Warning: package 'tidyr' was built under R version 4.4.3

Warning: package 'readr' was built under R version 4.4.3

Warning: package 'purrr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'stringr' was built under R version 4.4.3

Warning: package 'forcats' was built under R version 4.4.3
```

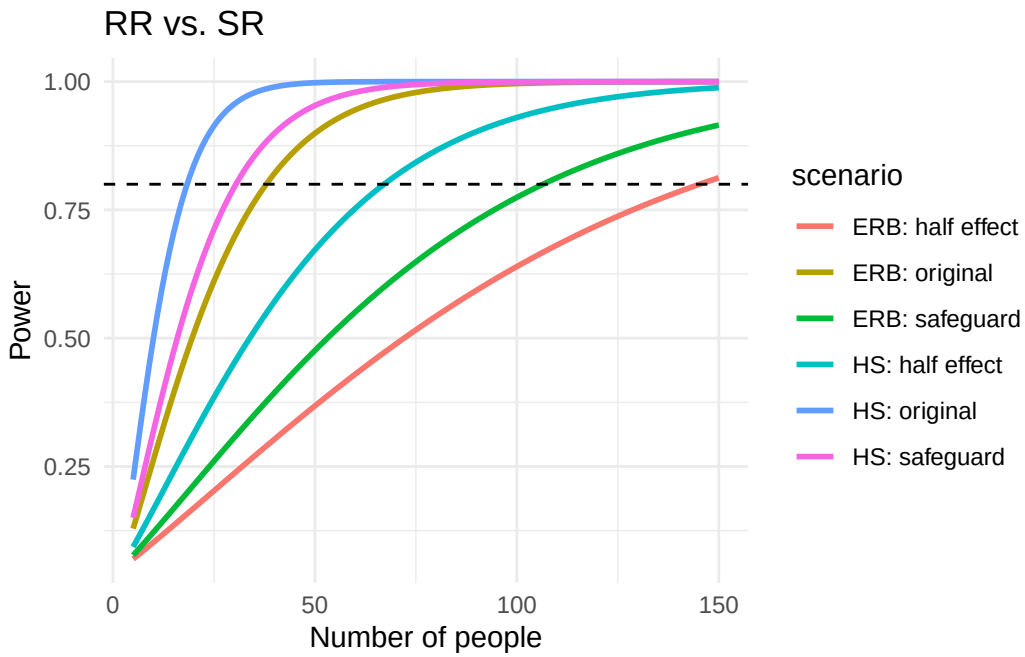
```
Warning: package 'lubridate' was built under R version 4.4.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
# effect sizes (d_z) from the planning step, for the RR vs. SR contrast
effects <- tibble(
  scenario = c("HS: original", "HS: half effect", "HS: safeguard",
               "ERB: original", "ERB: half effect", "ERB: safeguard"),
  d = c(0.693, 0.347, 0.525,
        0.467, 0.234, 0.274)
)

# power at each N (N = number of people, since the test is paired)
curve <- effects |>
  cross_join(tibble(n = 5:150)) |>
  mutate(power = map2_dbl(d, n, \(d, n) pwr.t.test(n = n, d = d, type = "paired")$power))

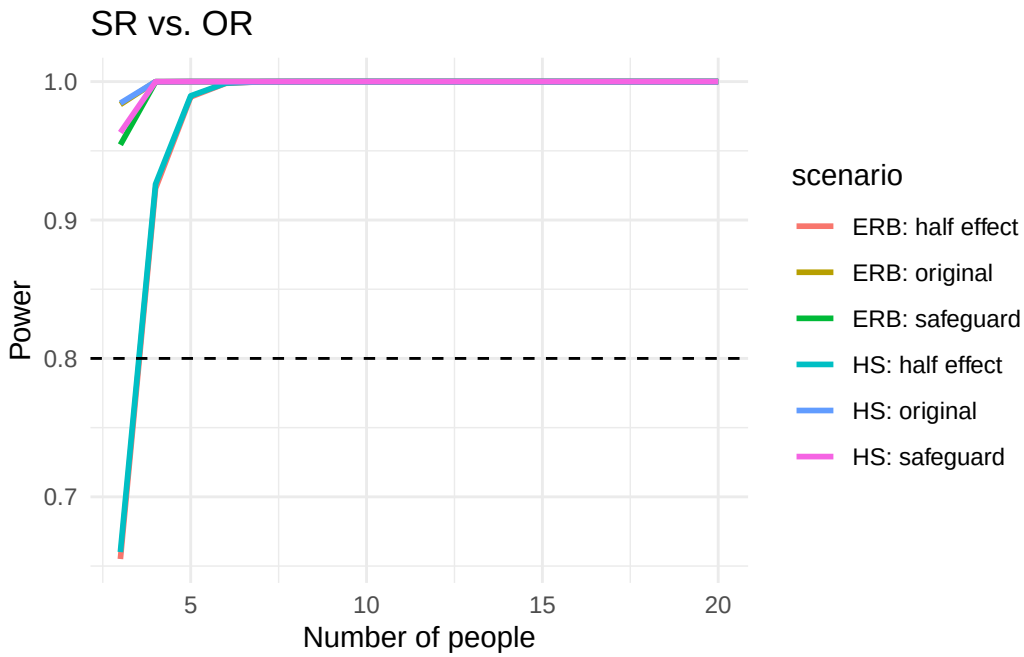
ggplot(curve, aes(n, power, colour = scenario)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = .8, linetype = "dashed") +
  labs(x = "Number of people", y = "Power", title = "RR vs. SR") +
  theme_minimal()
```




```
# effect sizes (d_z) from the planning step, for the SR vs. OR contrast
effects_or <- tibble(
  scenario = c("HS: original", "HS: half effect", "HS: safeguard",
               "ERB: original", "ERB: half effect", "ERB: safeguard"),
  d = c(5.302, 2.651, 4.715,
        5.263, 2.632, 4.556)
)

# the effect is huge, so power reaches 1 almost immediately: use a much smaller range of N
curve_or <- effects_or |>
  cross_join(tibble(n = 3:20)) |>
  mutate(power = map2_dbl(d, n, \(d, n) pwr.t.test(n = n, d = d, type = "paired")$power))

ggplot(curve_or, aes(n, power, colour = scenario)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = .8, linetype = "dashed") +
  labs(x = "Number of people", y = "Power", title = "SR vs. OR") +
  theme_minimal()
```




```
example_experiment <- function(n, effect) {
  treatment <- rnorm(n, mean = effect, sd = 1)
  control <- rnorm(n, mean = 0, sd = 1)
  t.test(treatment, control, var.equal = TRUE)$p.value
}

set.seed(91)
# HS vs. ERB overall accuracy: d = 0.64, 40 per group
mean(map_dbl(1:500, \(i) example_experiment(40, effect = 0.64)) < .05) # simulation
```

```
[1] 0.802
```

```
pwr.t.test(n = 40, d = 0.64)$power # formula
```

```
[1] 0.8069741
```

```
mean(map_dbl(1:500, \(i) example_experiment(40, effect = 0)) < .05) # should be near .05
```

```
[1] 0.056
```

```r
# within-person contrasts: one difference score per person
paired_experiment <- function(n, effect) {
  differences <- rnorm(n, mean = effect, sd = 1)
  t.test(differences)$p.value
}

set.seed(93)
# RR vs. SR, HS, half effect (0.347), 68 people
mean(map_dbl(1:500, \(i) paired_experiment(68, effect = 0.347)) < .05) # simulation
```

```
[1] 0.846
```

```r
pwr.t.test(n = 68, d = 0.347, type = "paired")$power # formula
```

```
[1] 0.8051704
```

```r
# SR vs. OR, HS, half effect (2.651), 5 people
mean(map_dbl(1:500, \(i) paired_experiment(5, effect = 2.651)) < .05)
```

```
[1] 0.986
```

```r
mean(map_dbl(1:500, \(i) paired_experiment(68, effect = 0)) < .05) # should be near .05
```

```
[1] 0.06
```

```r
interaction_experiment <- function(n, effect) {
  cells <- expand_grid(a = c(-0.5, 0.5), b = c(-0.5, 0.5), person = 1:n) |>
    mutate(y = effect * a * b + rnorm(n(), mean = 0, sd = 1))
  summary(lm(y ~ a * b, data = cells))$coefficients["a:b", "Pr(>|t|)"]
}

set.seed(92)
# group (HS vs. ERB) x RC type (RR vs. SR), 64 per cell, effect = 0.65
mean(map_dbl(1:500, \(i) interaction_experiment(64, effect = 0.65)) < .05)
```

```
[1] 0.726
```

```r
library(tidyverse)

one_experiment <- function(n, effect) {
  # 1. simulate one dataset: n people in total, each with one difference score
  #    (RR minus SR, or SR minus OR); the true mean difference is effect (in SD units)
  differences <- rnorm(n, mean = effect, sd = 1)
  # 2. confirmatory test: paired t-test, which is a one-sample t-test on the differences
  # 3. return the p-value
  t.test(differences)$p.value
}

set.seed(95)

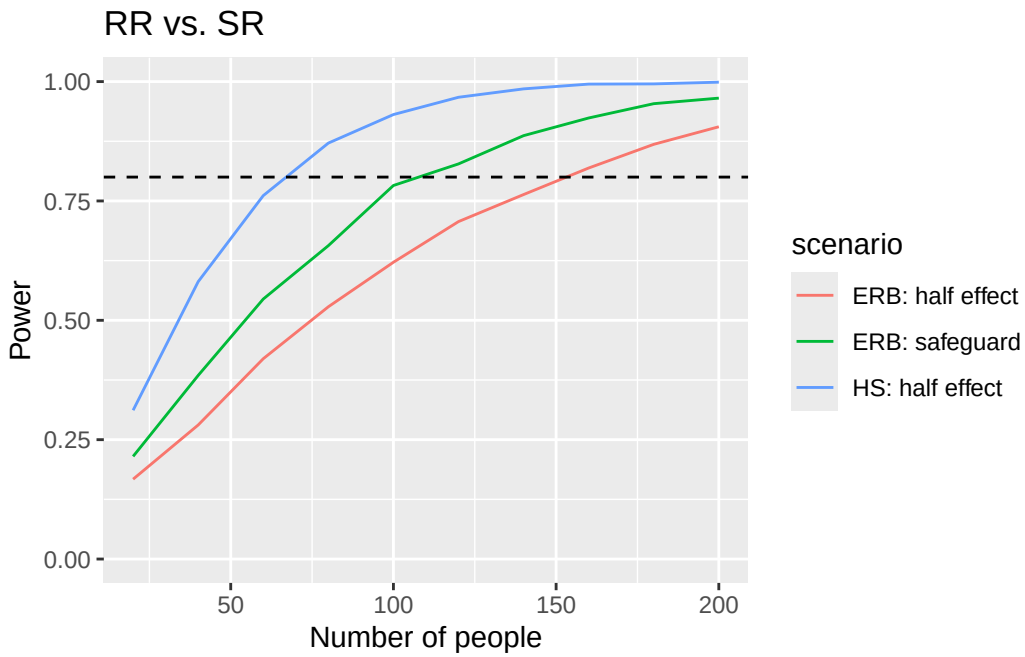
# sanity check: with no true effect, this should be near .05
mean(map_dbl(1:5000, \(i) one_experiment(80, effect = 0)) < .05)
```

```
[1] 0.0524
```

```r
# ---- RR vs. SR (the contrast that sets the sample size) ----
scenarios_rr_sr <- tibble(
  scenario = c("HS: half effect", "ERB: half effect", "ERB: safeguard"),
  effect   = c(0.35, 0.23, 0.27)
)

curve_rr_sr <- scenarios_rr_sr |>
  cross_join(tibble(n = seq(20, 200, by = 20))) |>
  mutate(power = map2_dbl(n, effect,
                          \(this_n, this_effect) mean(map_dbl(1:5000, \(i) one_experiment(th

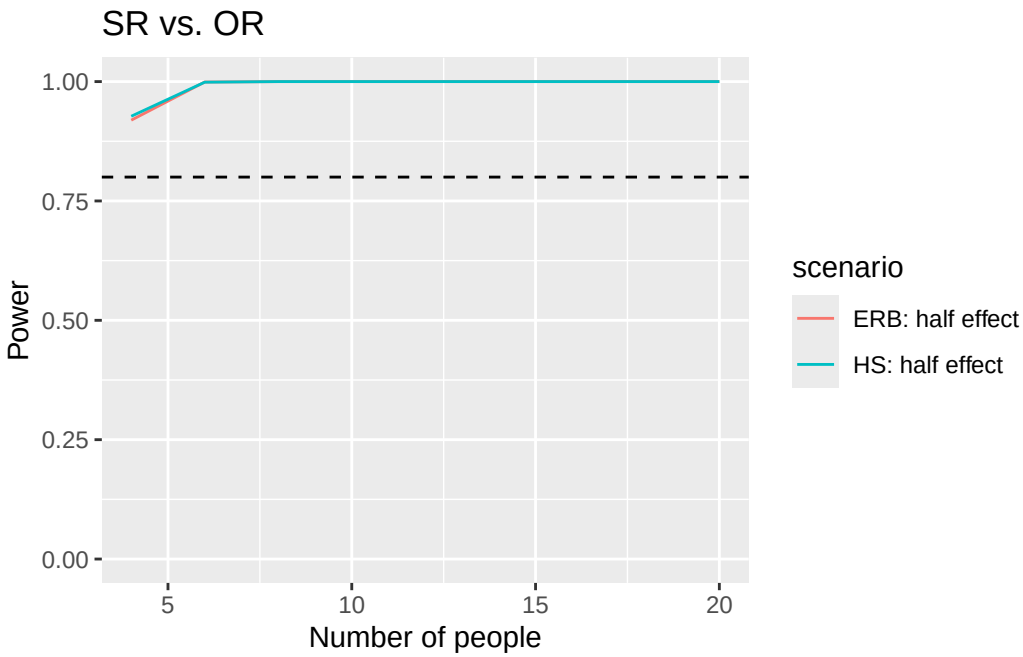
ggplot(curve_rr_sr, aes(x = n, y = power, colour = scenario)) +
  geom_line() +
  geom_hline(yintercept = .8, linetype = "dashed") +
  ylim(0, 1) +
  labs(title = "RR vs. SR", x = "Number of people", y = "Power")
```




```
# ---- SR vs. OR (huge effect, so a much smaller range of n) ----
scenarios_sr_or <- tibble(
  scenario = c("HS: half effect", "ERB: half effect"),
  effect   = c(2.65, 2.63)
)

curve_sr_or <- scenarios_sr_or |>
  cross_join(tibble(n = seq(4, 20, by = 2))) |>
  mutate(power = map2_dbl(n, effect,
                          \(this_n, this_effect) mean(map_dbl(1:5000, \(i) one_experiment(th

ggplot(curve_sr_or, aes(x = n, y = power, colour = scenario)) +
  geom_line() +
  geom_hline(yintercept = .8, linetype = "dashed") +
  ylim(0, 1) +
  labs(title = "SR vs. OR", x = "Number of people", y = "Power")
```



The original study (N = 31 HSs)reported a within-person SR vs. OR effect of d_Z = 5.30 (z=29.52). Because published effects tend to be inflated, we planned for half that effect, d_z=2.65, which needs only 4 participants for 80% power.Even a more conservative estimate, half of the effect implied by the log-odds coefficient (d=1.40), needs only 7. Sample size will therefore be set by feasibility and the analyses. To find the RR vs. SR effect, d_z = .69 (z = 3.86), I will run 78 HSs, which is about 2.5 times the original sample. This gives about 80% power to reject d33=.28, the effect the original study had 33% power to detect (Simonsohn, 2015). It also gives 80% power for effects of d_Z = .32 or larger. A non-significant result would tell us whether the effet is as larger as the original study could have detected, but not whether there is a smaller effect below about d_z =.3.

The original study (N = 21 ERBs)reported a within-person SR vs. OR effect of d_z = 5.26 (z = 24.12). Because published effects tend to be inflated, we planned for half that effect, d_z = 2.63, needs 4 participants for 80% power, and half of the effect implied by the log-odds coefficient (d = 0.91) needs 12. For the RR vs. SR d_z = .47 (z = 2.14), with a smaller and less reliable effect than for HSs. Planning for half that effect, d_z = .23, would need 146 participants for 80% power. This is more than I can recruit this quarter, so I will plan to run 53 ERBs which is 2.5 times the original sample. This gives about 80% power to reject d33 = .35, the effect the original study had 33% power to detect. It also gives 80% power for effects of d_z = .39 or larger. A non-significant result would tell us whether the effect is as large as the original study could have detected, but not whether there

is a smaller effect, below about d_z = .4.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

The planned study will collect heritage Spanish-English bilinguals and Second language learners of Spanish from the SONA participant pool. This pool will contain college students, aged 18 years or older.

Data exclusions will be applied to any participants that have Spanish listed as their native language and have grown up in a Spanish dominant environment past childhood. Any participants that score low on the RR RC-type will be excluded as that will be indication of not understanding the task or not paying attention. Reaction time will be excluded as well however different from the original paper, since reaction time on JsPsych begins at the onset of audio. Reaction time faster than 150 ms will be excluded and slower than 10000 after audio offset will be excluded.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Picture-Matching Task:

Fuchs and colleagues (2026) tested whether heritage speakers' relative amount of cognitive effort required for Spanish language processing. To do so, they recruited two additional populations for comparison against heritage speakers: Spanish-raised bilinguals (SRBs) and late second-language learners (English-rasied bilinguals, ERBs). The study consisted of three different tasks, a single picture-matching task, a non-linguistic task, and a dual-task combining the linguistic and non-linguistic task. For the purpose of this replication project, I will be replicating the linguistic task: A picture-matching task where "participants matched candidate images with auditorily presented linguistic stimuli" The linguistic picture-matching task contained 120 trials which had "120 trials. These were all Spanish phrases, heard in the presence of two images." Additionally, "The phrase was always of the form ARTICLE + NOUN + RELATIVE CLAUSE. Phrase difficulty was manipulated by increasing the complexity of relative clauses, from Reduced Relative (RR) to Subject Relative (SR) to Object Relative (OR), as illustrated in (1). (1) RR: el oso pintando el tigre "the bear painting the tiger" SR: el oso que pinta al tigre "the bear that is painting the tiger" OR: elosoal quepinta el tigre "the bear that the tiger is painting"

Listeners are asked to match the phrase they heard to the correct image by either selecting left-arrow for the left image or right-arrow for the right image. "There were 20 total animals and 20 action verbs combined to create 80 different animal-action pairs. Audio stimuli for the task were recorded in a sound-isolated booth by a first-language speaker of Spanish with a Mexican dialect. All audio was recorded at 44.1 kHz and presented monaurally using speakers; the average acoustic intensity of the final 240 audio stimuli was 0.004 mean squared amplitude (SD = 0.00001) (equivalent to 24.1 decibels relative to full scale)."

Additionally the picture-matching task "were blocked into sections of 30 trials, with an optional break between each block. Instructions for the two single tasks were provided at the onset of the experiment."

The study does not mention how many sentence structures are shown per trial but since the materials include 40 RR, 40 SR, and 40 OR audio files, it will be split across 4 sections of 30 trials and no audio will be repeated. This ensures that all participants have been exposed to the same conditions however order of trials will be randomized across each participant.

Language-history questionnaire:

To characterize the population, prior to the task, participants will receive a language background questionnaire. The paper states that they used a "a pen-and-paper version ofthe Bilingual Language Profile (BLP) (Birdsong et al., 2012) and an additional brief questionnaire consisting of selected questions from the Language Experience and Proficiency Questionnaire(LEAP-Q)(Kaushanskaya et al., 2020; Marian et al., 2007)." However, due to this experiment using JsPysch, the questionnaire will be administered online. The important questions to assess heritage speakers of Spanish in this study is including questions regarding age of language acquisition for Spanish and English and place of birth/time of immigration. Heritage speakers are characterized as have acquired the heritage language at birth and English later (first year of schooling) and were born and raised in the US or immigrated to the US prior to or at the age of 6. Second language learners of Spanish are those who have enrolled in Spanish courses in high school or University.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

This study was administered in a "private space and followed the same procedure. Participants first gave informed consent for participation in the study. They then completed the Flanker task and the receptive vocabulary task. Following this, they completed the main experimental task, which consisted of blocked single-task trials and then blocked dual-task trials. Within each block, participants initiated each trial with a key press, meaning inter-stimulus intervals were self-timed; breaks between experimental blocks were explicitly signaled and self-timed."

The study was also completed on a laptop in the laboratory and ended the experiment with completing the language background questionnaire on paper. All participants were university students.

Difference: In this study, it will be completed fully online and administered through SONA: online experiment and subject pool management software. The questionnaire and linguistic task are designed and administered through JsPsych. Participants will receive compensation in the form of SONA credits which can be applied as extra-credit for their applicable university courses.

Participants will first completed a language-history questionnaire then directed to the picture-matching task in which all instructions will be given on the screen. An audio check is added to ensure participants can properly hear the audios.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

In this study, the authors "took a general mixed effects modelin gapproach for analysis. All data were processed and analyzed in R Version 4.2.1(R CoreTeam, 2021) using the lme4 package (Bates et al., 2015). For each analysis,the maximal random effect structure supported by the data was included; if a model including random slopes and intercepts did not converge, the random effects structure was reduced until convergence was reached. Random intercepts grouped by participant were prioritized over random intercepts grouped by item when a model with both of these did not converge, to account for typically larger by-participant than by-item variation (Sonderegger, 2023)." This is to check if the difficulty manipulation worked and that performance on OR will show less accurate responses compared to the other two sentence structures.

For the picture-matching task, "responses faster than 150 ms and outside of two standard deviations of the mean (grouped by participant and by relative clause type) were excluded from analysis, removing 2.8% of the data, at approximately equal rates across the three groups and across conditions within each group."

The planned confirmatory test for this will be a logistic mixed-effects regression on trial-level accuracy that compares the performance on the RC-type conditions. This can be one with a Wald z-test on each condition across the heritage speakers. Specifically comparing SR vs. OR type, if coefficient is 0, then there was no accuracy different between the two. If coefficient is positive and significant then SR is more accurate than OR which replicates the findings of the original paper.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

The key analysis of interest here is performance on the RC type specifically looking at the difference of performance between SRs and ORs. This is because the original study found a significant effect of RC type on SRs than ORs (=3.27,SE=0.07,z=44.14,p<0.001). Similar to the original paper, I will use a "logistic mixed effects regression model was fitted to the data,predicting log-odds of the probability of a correct response by group,RC type and their interaction,as well as random intercepts grouped by item (models with random effects grouped by participant did not converge)."

For this study, I will implement a similar analysis plan with a focus on RC-type slopes and by-item random intercepts. If enough data collected across different participants groups, then another analysis by participant type will also be conducted.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

The known differences from the original study is first the procedure. In the original study, participants first completed a receptive vocabulary task prior to the picture-matching task to ensure their familiarity with all items. However, in this replication, participants will be assumed to have enough Spanish knowledge to recognize all items. Additionally, since the original study reported that all participants scored high, this will be considered the norming of the items used to test a broader population.

The entire study will also be administered online through SONA and using JsPsych and hosted on GitHub Pages. This is different as the original study was conducted in the lab. To control for any environmental noise, audio checks and attention checks will be added.

The heritage Spanish-English population should be a similar sample to the one in the original study however recruiting second-language Spanish learners will be more broad and not from students who are currently enrolled in university-level courses in Spanish.

Additionally, participants will not be given the flanker task prior to the picture-matching task and instead of completing the questionnaire at the end of the experiment, they will be completing it in the beginning prior to the task. This is to ensure that participants' answers on the proficiency questions are not influenced by their performance on the task.

The analysis plan and exclusion criteria will look different since I do not have three different types of group and will look at the RC-type group instead. Additionally, on JsPsych reaction time will start from the onset of the audio to the response therefore my exclusion for reaction time should be different. Reaction time will be measured by the onset to the end of the audio.

Any participants that score low on the RR condition will also be excluded as it is considered the baseline and means that participant did not understand the task or is not paying attention.

The study's main research question was to test whether heritage speakers' of Spanish are engaging in more cognitive effort when they are required to process language in Spanish. They tested this by comparing them to two other groups and in a dual-task that includes both a linguisitc and non-linguistic task. However, the purpose of this replication is to test the claim of relative-clause processing and comprehension in heritage speakers and how there is an asymmetry between SRs and ORs due to what has been identified in the literature as a hierarchy of processing difficulty. If processing cost is greater for object-extracted relative clauses for the heritage speakers of this sample, I should be able to replicate the findings of the original study.

I do not believe there are any major differences between my replication and the original paper as I am using the same type of stimuli all provided by the authors of the original paper. However, since I do not have the same group of participants, comparisons across groups will not look the same and my analysis will rely more on the accuracy across the RC-type condition. This is still an important test because the main claim of the paper is with increase syntactic complexity, the more cognitive resources that will be employed leading to a lower performance on the dual-task conducted later in the paper.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

```
# The original tried these random-effects structures in this order (singular / failed to conv
# kept the first model that worked, which in their data was the item-intercept model. This he
# it takes the first formula that fits without any warning and without a singular fit.
# it takes the first formula that fits without any warning and without a singular fit.
# TRUE if every grouping factor of the random effects has at least 2 levels in the data
# (a term such as (1|subject) cannot be fitted when the data contain only ONE participant)

grouping_ok <- function(f, data) {
  bars <- lme4::findbars(f)
  all(vapply(bars, function(bar) length(unique(data[[deparse(bar[[3]])]])) > 1, logical(1)))
}
# fit a given formula: mixed model if it has random effects, ordinary logistic regression if
fit_given <- function(f, data) {
  if (length(lme4::findbars(f)) == 0) glm(f, data = data, family = binomial) else glmer(f, da
}
fit_first_ok <- function(formulas, data) {
  ok <- vapply(formulas, grouping_ok, logical(1), data = data)
  usable <- formulas[ok]
  for (f in formulas[!ok]) cat("  skipped (a grouping factor has only one level, e.g. one par
  for (f in usable) {
    fit <- tryCatch(glmer(f, data = data, family = binomial), error = function(e) NULL, warni
    if (!is.null(fit) && !isSingular(fit)) { cat("  random-effects structure kept:", deparse1
    cat("  not usable (error / failed to converge / singular):", deparse1(f), "\n")
  }
  for (f in usable) {                          # nothing fitted cleanly: use the first struct
    fit <- tryCatch(suppressWarnings(glmer(f, data = data, family = binomial)), error = funct
    if (!is.null(fit)) {
      warning("No structure fitted cleanly; using ", deparse1(f), ". Interpret with caution."
      return(list(fit = fit, formula = f))
    }
  }
  f0 <- lme4::nobars(formulas[[1]])            # last resort: no random effects possible with
  warning("Too few participants/items for any random effect; using an ordinary logistic regre
  list(fit = glm(f0, data = data, family = binomial), formula = f0)
}

if (n_groups == 2) {
  # logistic regression models
  cat("\n##### MAIN MODEL: aud_corr ~ group*aud_diff #####\n")
```

```
  main <- fit_first_ok(list(
    aud_corr ~ group*aud_diff + (1+aud_diff|subject),
    aud_corr ~ group*aud_diff + (1+aud_diff||subject),
    aud_corr ~ group*aud_diff + (1|subject),
    aud_corr ~ group*aud_diff + (1|audiostimulus)), LING_foranalysis)
  aud_glm1 <- main$fit
  print(summary(aud_glm1))

  # relevel to check effect of L2 (same random-effects structure as the main model)
  cat("\n##### RELEVELED (group_relevel: contrast1 = L2) #####\n")
  relevel_formula <- as.formula(gsub("\\bgroup\\b", "group_relevel", deparse1(main$formula)))
  aud_glm1_relevel <- fit_given(relevel_formula, LING_foranalysis)
  print(summary(aud_glm1_relevel))

  # the original found an interaction between group and RC type (aud_diff), so it unpacked ea
  sub_formulas <- list(aud_corr ~ aud_diff + (1|audiostimulus),
                       aud_corr ~ aud_diff + (1|subject),
                       aud_corr ~ aud_diff + (1|audiostimulus) + (1|subject))
  # HSs
  cat("\n##### HSs #####\n")
  aud_glm1_HS <- fit_first_ok(sub_formulas, LING_foranalysis[LING_foranalysis$group=="HS",])$
  print(summary(aud_glm1_HS))   # contrasts: RR vs SR, SR vs OR

  # L2s (= ERBs)
  cat("\n##### L2s (ERBs) #####\n")
  aud_glm1_L2 <- fit_first_ok(sub_formulas, LING_foranalysis[LING_foranalysis$group=="L2",])$
  print(summary(aud_glm1_L2))

  # save the four summaries to a text file (capture.output is safe inside knitr/Quarto, unli
  capture.output(print(summary(aud_glm1)), print(summary(aud_glm1_relevel)), print(summary(au
                 file = file.path(out_dir, "model_summaries.txt"))
} else {
  cat("\nOnly", n_groups, "group found in the data. The group-by-RC-type models need both HS
      "so the models are skipped; the plot below still works.\n")
}
```

```
##### MAIN MODEL: aud_corr ~ group*aud_diff #####

Warning: the 'findbars' function has moved to the reformulas package. Please
update your imports, or ask an upstream package maintainer to do so.
```

```
  not usable (error / failed to converge / singular): aud_corr ~ group * aud_diff + (1 + aud

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ group * aud_diff + (1 + aud

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ group * aud_diff + (1 | sub

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ group * aud_diff + (1 | aud

Warning in fit_first_ok(list(aud_corr ~ group * aud_diff + (1 + aud_diff | : No
structure fitted cleanly; using aud_corr ~ group * aud_diff + (1 + aud_diff |
subject). Interpret with caution.

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: aud_corr ~ group * aud_diff + (1 + aud_diff | subject)
   Data: data

      AIC       BIC    logLik -2*log(L)  df.resid
    486.0     540.6    -231.0     462.0       687

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.3072 -0.1869  0.1245  0.3705  5.3506

Random effects:
 Groups  Name        Variance Std.Dev. Corr
 subject (Intercept)  0.2563  0.5062
         aud_diff1    3.5399  1.8815   -0.94
         aud_diff2   17.8944  4.2302    0.99 -0.97
Number of obs: 699, groups:  subject, 6

Fixed effects:
                Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept)        1.5598     0.4161   3.748 0.000178 ***
group1             0.4402     0.3355   1.312 0.189563
aud_diff1         -1.8740     1.3846  -1.353 0.175901
aud_diff2          7.0837     2.3140   3.061 0.002204 **
group1:aud_diff1  -0.8375     1.1640  -0.720 0.471821
group1:aud_diff2   2.9251     2.1697   1.348 0.177599
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) group1 ad_df1 ad_df2 gr1:_1
group1       0.579
aud_diff1   -0.779 -0.419
aud_diff2    0.769  0.428 -0.883
grp1:d_dff1 -0.406 -0.742  0.577 -0.496
grp1:d_dff2  0.367  0.803 -0.441  0.495 -0.897
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.0344117 (tol = 0.002, component 1)
  See ?lme4::convergence and ?lme4::troubleshooting.


##### RELEVELED (group_relevel: contrast1 = L2) #####

Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed t
  See ?lme4::convergence and ?lme4::troubleshooting.

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: aud_corr ~ group_relevel * aud_diff + (1 + aud_diff | subject)
   Data: data

      AIC       BIC    logLik -2*log(L)  df.resid
    486.0     540.6    -231.0     462.0       687

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.3317 -0.1867  0.1239  0.3708  5.3571

Random effects:
 Groups  Name        Variance Std.Dev. Corr
 subject (Intercept)  0.2637  0.5135
```

```
         aud_diff1    3.6446  1.9091   -0.94
         aud_diff2   18.2678  4.2741    0.99 -0.97
Number of obs: 699, groups:  subject, 6

Fixed effects:
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)                1.5668     0.4210   3.722 0.000198 ***
group_relevel1            -0.4484     0.3401  -1.318 0.187393
aud_diff1                 -1.8903     1.4026  -1.348 0.177766
aud_diff2                  7.1052     2.3416   3.034 0.002410 **
group_relevel1:aud_diff1   0.8586     1.1823   0.726 0.467730
group_relevel1:aud_diff2  -2.9673     2.2006  -1.348 0.177517
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) grp_r1 ad_df1 ad_df2 g_1:_1
group_rlvl1 -0.586
aud_diff1   -0.783  0.428
aud_diff2    0.773 -0.434 -0.885
grp_rlv1:_1  0.415 -0.747 -0.583  0.502
grp_rlv1:_2 -0.377  0.807  0.449 -0.500 -0.900
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.0443163 (tol = 0.002, component 1)
  See ?lme4::convergence and ?lme4::troubleshooting.


##### HSs #####

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ aud_diff + (1 | audiostimulu

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ aud_diff + (1 | subject)

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ aud_diff + (1 | audiostimulu
```

```
boundary (singular) fit: see help('isSingular')

Warning in fit_first_ok(sub_formulas, LING_foranalysis[LING_foranalysis$group
== : No structure fitted cleanly; using aud_corr ~ aud_diff + (1 |
audiostimulus). Interpret with caution.

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: aud_corr ~ aud_diff + (1 | audiostimulus)
   Data: data

      AIC       BIC    logLik -2*log(L)  df.resid
     81.3      95.1     -36.6      73.3       229

Scaled residuals:
    Min      1Q  Median      3Q     Max
-8.7178 -0.2617  0.1147  0.2000  3.8210

Random effects:
 Groups        Name        Variance  Std.Dev.
 audiostimulus (Intercept) 2.778e-14 1.667e-07
Number of obs: 233, groups:  audiostimulus, 175

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.6229     0.4181   3.881 0.000104 ***
aud_diff1    -1.1119     1.1661  -0.953 0.340351
aud_diff2     7.0118     1.1076   6.330 2.45e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
          (Intr) ad_df1
aud_diff1 -0.456
aud_diff2  0.575 -0.784
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')


##### L2s (ERBs) #####
```

```
boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ aud_diff + (1 | audiostimulu

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ aud_diff + (1 | subject)

boundary (singular) fit: see help('isSingular')

  not usable (error / failed to converge / singular): aud_corr ~ aud_diff + (1 | audiostimulu

boundary (singular) fit: see help('isSingular')

Warning in fit_first_ok(sub_formulas, LING_foranalysis[LING_foranalysis$group
== : No structure fitted cleanly; using aud_corr ~ aud_diff + (1 |
audiostimulus). Interpret with caution.

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: aud_corr ~ aud_diff + (1 | audiostimulus)
   Data: data

      AIC       BIC    logLik -2*log(L)  df.resid
    484.0     500.6    -238.0     476.0       462

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.2803 -0.6667  0.4385  0.4695  1.5000

Random effects:
 Groups        Name        Variance Std.Dev.
 audiostimulus (Intercept) 0        0
Number of obs: 466, groups:  audiostimulus, 220

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.7832     0.1161   6.745 1.54e-11 ***
aud_diff1     0.1367     0.3021   0.452    0.651
```

```
aud_diff2     2.3229     0.2714   8.558  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
          (Intr) ad_df1
aud_diff1  0.039
aud_diff2  0.143 -0.531
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

```
## ==================================
## PICTURE-MATCHING ACCURACY PLOT
## ==================================

# create new table with accuracies per participant per condition
audio_accuracy <- single_audio_clean %>%
  group_by(group, subject, aud_diff) %>%
  summarise(accuracy=mean(aud_corr))
```

```
`summarise()` has grouped output by 'group', 'subject'. You can override using
the `.groups` argument.
```

```
# Get an overall barplot from this
aud_sum <- audio_accuracy %>% group_by(group, aud_diff) %>%
  summarize(mean_acc=mean(accuracy),
            se=sqrt(mean(accuracy)*(1-mean(accuracy))/length(accuracy)))
```

```
`summarise()` has grouped output by 'group'. You can override using the
`.groups` argument.
```

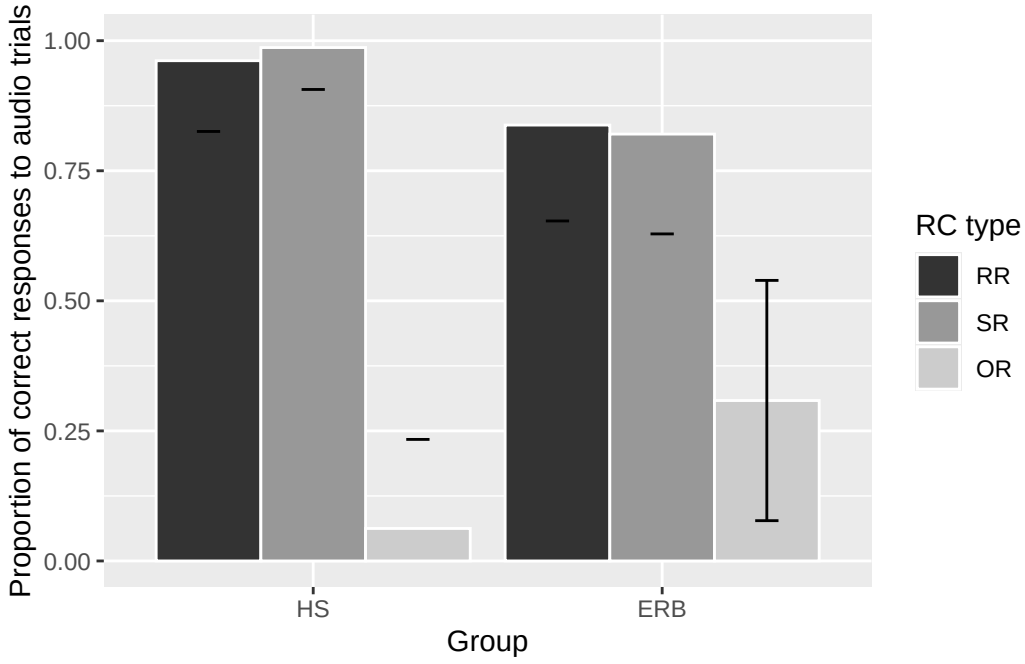
```
print(aud_sum)
```

```
# A tibble: 6 x 4
# Groups:   group [2]
  group aud_diff mean_acc     se
  <chr> <fct>       <dbl>  <dbl>
1 HS    RR          0.962  0.136
2 HS    SR          0.987  0.0806
3 HS    OR          0.0625 0.171
```

```
4 L2    RR          0.838  0.184
5 L2    SR          0.821  0.192
6 L2    OR          0.308  0.231
```

```r
dualFig3 <- aud_sum %>% filter(group!= "Gen1.5") %>% ggplot(aes(x=group,y=mean_acc,fill=aud_c
  geom_bar(stat="identity", color="white",position=position_dodge()) +
  geom_errorbar(aes(ymin=mean_acc-se, ymax=mean_acc+se),width=0.2,position=position_dodge(0.9
  ylim(c(0,1)) +
  ylab("Proportion of correct responses to audio trials") + xlab("Group") +
  scale_fill_grey() + labs(fill="RC type") +
  scale_x_discrete(labels = c("HS" = "HS",
                              "L2" = "ERB"))     # CHANGED: no SRB group in your data

ggsave(
  filename = file.path(out_dir, "Replication-Fig3.png"),
  plot = dualFig3,
  width = 6,          # width in inches
  height = 4,         # height in inches
  dpi = 600           # resolution
)
print(dualFig3)    # shows the plot
```



Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.